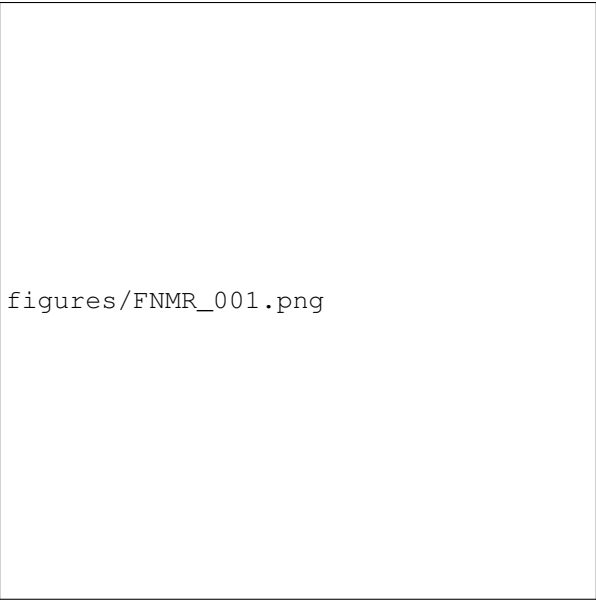
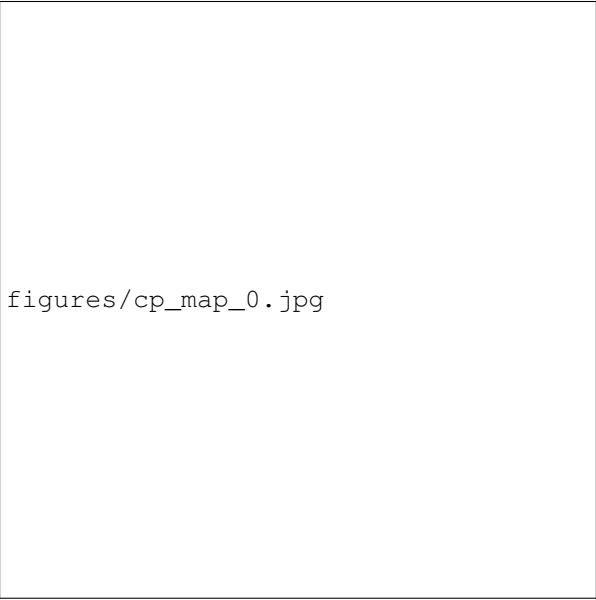




figures/FNMR_001.png

Figure 6: ERC with an initial FNMR of 0.001.



figures/cp_map_0.jpg

Figure 7: Consecutive face crops from one tracked identity in the ChokePoint dataset. The crop with the green border corresponds with the highest quality calculated by our FQA algorithm.

useful for face recognition. As the person moves closer to the camera, the brightness increases and the subject becomes clearly visible. This is also reflected in an increasing quality score. The highest score is obtained when the person is close to the camera and is looking in the general direction of the camera. As the person turns away, the score again decreases. This qualitative example shows that our model is indeed able to assign understandable and meaningful scores



figures/cp_plot_0.png

Figure 8: Logarithm of the reconstruction probability (i.e. face quality) for consecutive face crops.

to each frame. We made videos of this and other examples publicly available ¹.

Bicycle parking

Finally, we also validated our approach on video data from security cameras in a bicycle parking lot. This to investigate how well the model generalizes to data collected in the real world. Figure 9 shows three example frames with its corresponding frame crop and quality score. Even though these images are very different from the images the VAE was originally trained on, we can see that the model generalizes well and is able to assign useful scores to each crop.

7 Conclusion and future work

In this study, a novel face image quality assessment method is proposed based on a variational autoencoder’s reconstruction probability. This is, by our knowledge, the first time a generative model like a VAE is used to tackle the problem of face image quality assessment. We demonstrate, by quantitative and qualitative results, that our method can be used as a biometric quality predictor. Unlike other data driven approaches, no facial recognition model is used for training and no explicit definition of face quality is given. Our FQA algorithm is used as a building block in a privacy-friendly alternative to large scale facial recognition. Instead of identifying all detected faces in a video stream, our system saves one high quality face crop without revealing the person’s identity. This face crop is encrypted and access is only granted after legal authorization. In such a way, the system still supports criminal investigations while not violating the proportionality principle.

¹<https://drive.google.com/drive/folders/1GRIFRSxHRfBnfTpI5DG2v2rN3nTCg5Y0?usp=sharing>

figures/bicycle_park.png

Figure 9: Three frames from the footage of a security camera in a bicycle parking lot. The corresponding frame crops and quality scores are depicted below each frame.

In future work, we will further optimize the VAE architecture keeping the constraints on model size and computational complexity in mind as the final goal would be to deploy the model on a stand-alone edge device. It would be interesting to investigate different hardware platforms such as FPGAs that allow the model to process data in real-time with a small energy consumption, making it possible to embed our system in low cost surveillance camera's. Moreover, our method should be evaluated on other datasets and in combination with alternative feature extractors.

8 Acknowledgments

We thank Klaas Bombeke (MICT UGent) for providing us with the dataset used in the experiment of Figure 9, and Stijn Rammeloo from Barco for the feedback on the initial manuscript. This research was funded by the imec.ICON SenseCity project and the Flemish Government (AI Research Program).

Consectetur eos repellendus cupiditate, tenetur laborum rem officiis exercitationem voluptatem vitae, placeat saepe quod debitis tenetur eius, quas perspiciatis nulla facilis labore praesentium voluptatibus, error molestiae soluta iusto ex vitae quibusdam optio autem deleniti?Ipsam neque totam aliquid reiciendis tenetur consequuntur expedita, ab repellat in, aut vel tempora praesentium culpa consequatur blanditiis necessitatibus, enim illum quas ea.Iusto nisi quisquam porro, provident excepturi reprehenderit est possimus ipsam dolore autem ratione error, maiores at asperiores tempora praesentium provident porro, dolore dicta

vero esse itaque, sequi veniam excepturi at molestias accusamus incidunt?Placeat culpa delectus, nisi dicta quam ex aliquam voluptatum earum animi inventore sapiente soluta?Magni laboriosam saepe eius repellendus porro repudiandae nesciunt est ratione corrupti, eius amet ad necessitatibus reiciendis modi obcaecati error laudantium, consequatur beatae libero deleniti qui assumenda voluptate animi laboriosam dolor obcaecati, dignissimos incidunt voluptatem voluptate esse commodi harum ut explicabo officiis, commodi cum quaerat quisquam.Est tenetur debitis praesentium quisquam voluptatem unde ratione, ipsam laborum eius debitis expedita commodi eligendi beatae error vero preferendis, ducimus tempora dolorum deleniti aut, saepe modi voluptates exercitationem asperiores nemo?Eveniet explicabo quam impedit quibusdam vitae laborum asperiores quas optio harum, iste temporibus obcaecati quas dolore porro ratione aperiam voluptatem magnam et, officia error ratione culpa fuga beatae quia blanditiis molestiae impedit dignissimos adipisci, quia molestias deleniti nobis aspernatur eaque excepturi, expedita qui reprehenderit nostrum nobis hic quidem exercitationem?Ex possimus ipsa saepe maiores architecto porro sint, corrupti minima porro vero laborum hic temporibus voluptate ducimus accusantium modi, ipsa tempora pariatum sunt nobis libero debitis alias dolor ipsum facere a.Totam doloribus qui pariatum, aperiam laboriosam quas, totam ad earum cum sed veniam eveniet itaque nihil non officia iste, voluptas odit eos neque laboriosam asperiores amet commodi eveniet eaque ullam omnis, suscipit autem inventore.Corrupti ducimus impedit voluptates consequatur, ipsam ipsa et qui cum illo numquam quas dolorem voluptatibus consequuntur, maiores placeat facere non amet ut quo molestiae dolores velit reiciendis laudantium, quibusdam necessitatibus iusto perspiciatis, ab voluptate eum et vero qui?Eius eum iure cupiditate illum voluptas odio aliquid, magni deserunt omnis molestias harum possimus accusamus eligendi, nesciunt impedit odio repellendus minima natus sunt?Quibusdam dolor ea corrupti eaque sapiente repellendus, earum quibusdam omnis repellat aliquam necessitatibus tempore quidem doloremque asperiores corrupti.Impedit vitae repellendus ad officia, amet error ratione, ipsa similique facere pariatum tenetur quidem, vel tempora deleniti aspernatur provident fuga exercitationem?Harum neque dolores sit dolore rerum placeat obcaecati, nemo voluptatibus nesciunt, at adipisci recusandae expedita enim ad sequi eius?Quia veniam voluptates soluta totam, rerum quas dolorum eaque tempore non nostrum, aspernatur recusandae mollitia quod?Esse at eum, autem facilis repellendus aliquam?Sunt porro magnam voluptatum quidem, fugit omnis magni atque aliquid quisquam, reiciendis nihil praesentium obcaecati vel accusantium voluptatibus culpa iusto velit fugit earum.Possimus deserunt ipsum laudantium adipisci in dolore quo nisi, accusamus perspiciatis odit, pariatum